Campus Recruitment Training Phase 2

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Sem : 6
Batch - B1
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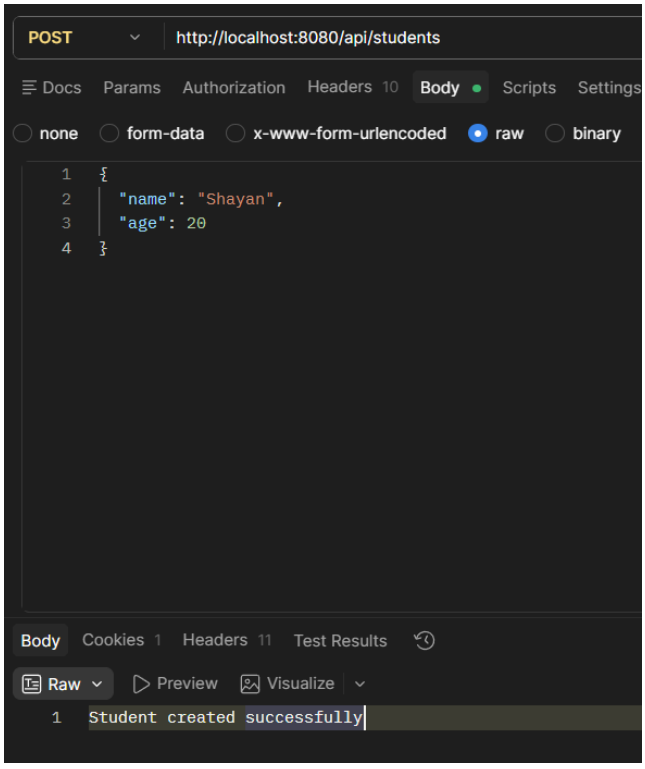
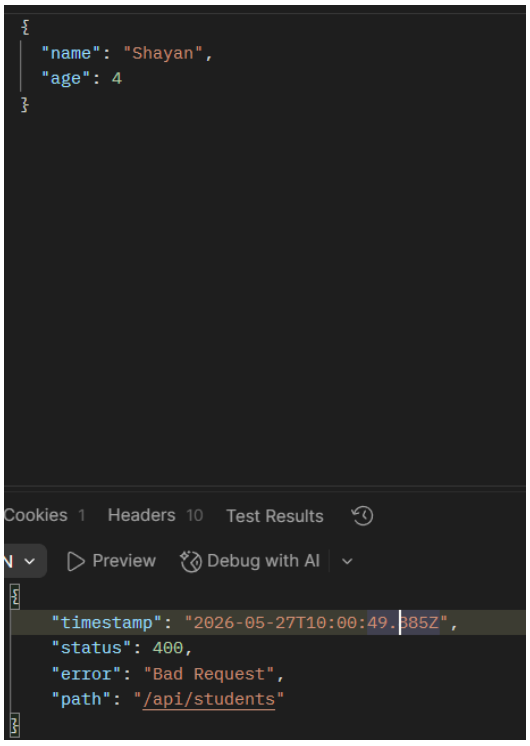
Q1)
Output
Constructor called
Init called

Q2)
Corrected code:
return new ResponseEntity<>(u, HttpStatus.OK);

Q3)





Q4)
Annotation : @Transactional
Both credit and debit are treated as single operation , either they execute successfully or fail , there is no inconsistent state.

Q6)
Corrected code:
String token = header.substring(7);

Filter validates jwt request header

Q8)
Problem: N+1 query problem occurs when
1 query loads parent entities and N queries load related child entities.
Cause: Lazy loading inside loops.
Fix: Use JOIN FETCH in JPQL to load relationships in one query.

Q11)
Fixed code:
@Query("SELECT o FROM Order o JOIN FETCH o.items")

Q16)
Benefits of splitting into microservices:

1. Each module or service can be deployed and scaled separately
2. Crash of one service doesnt affect others
3. We can use separate tech stack for each service and improve its performance

Challenges

1. Increased complexity
2. Harder debugging and monitoring because logs and flows are distributed across multiple services.

(c) Eureka

Eureka acts as a **service registry** where all microservices (User, Order, Product) register themselves with their service names

The API Gateway acts as a **single entry point** for the React frontend, which forwards incoming requests to the correct service by looking up their locations in Eureka dynamically.